

- 3 The variation with displacement x of the acceleration a of a 5.0 g particle is shown in Fig. 3.1.

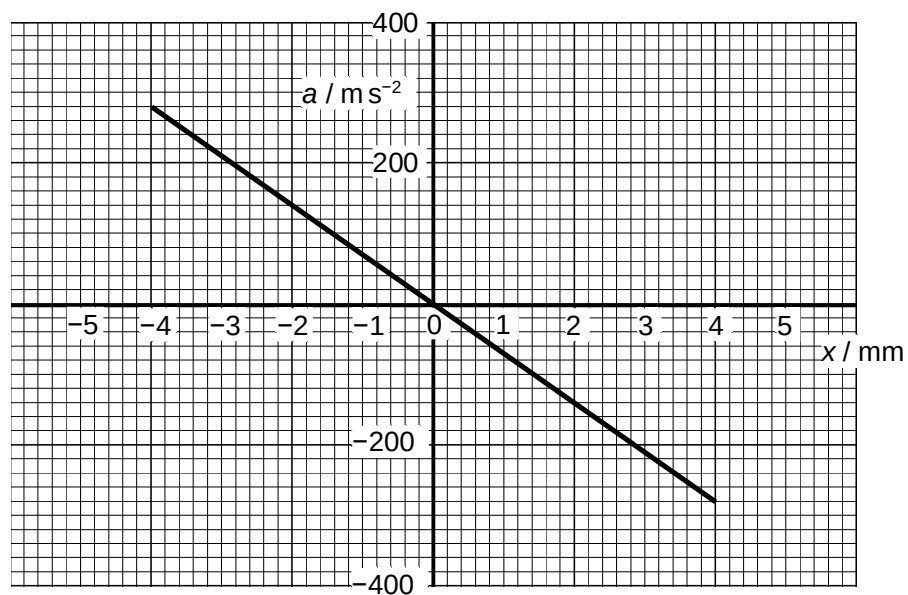


Fig. 3.1

- (a) Explain how it may be deduced from the graph that the oscillations of the particle are simple harmonic.

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..... [2]

- (b) Calculate the frequency of oscillations of the particle.

frequency = Hz [3]

(c) Determine the maximum speed of the particle.

maximum speed = m s^{-1} [2]

(d) Calculate the total energy of the particle.

total energy = J [2]

10

4 (a) A 50 W transmitter emits microwaves of frequency 10 GHz in all directions, as shown